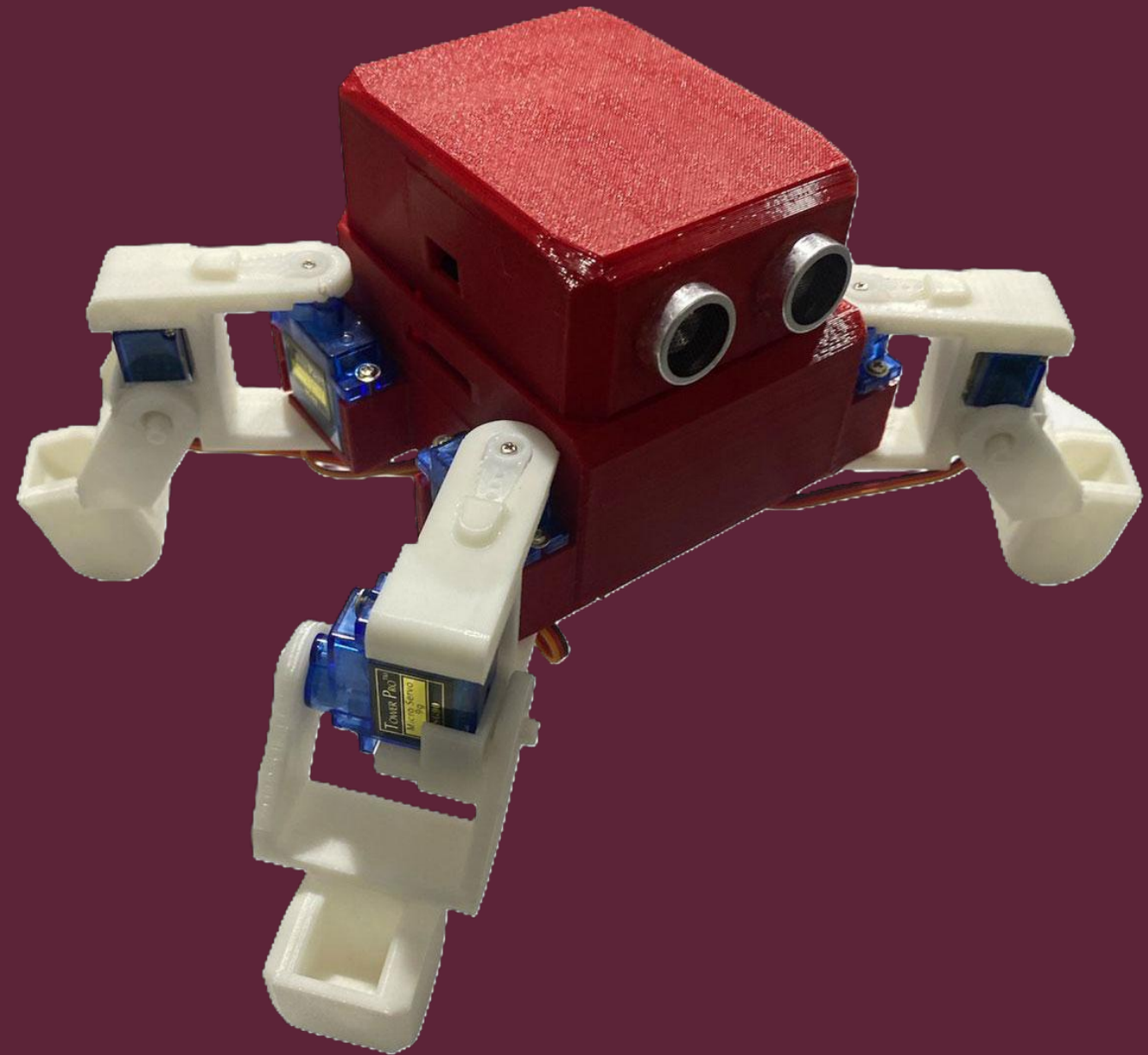
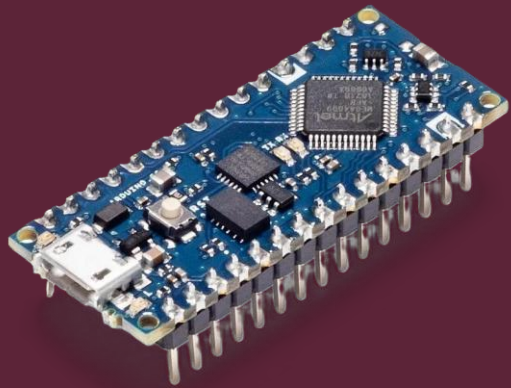


Otto — Quadruped Project Overview

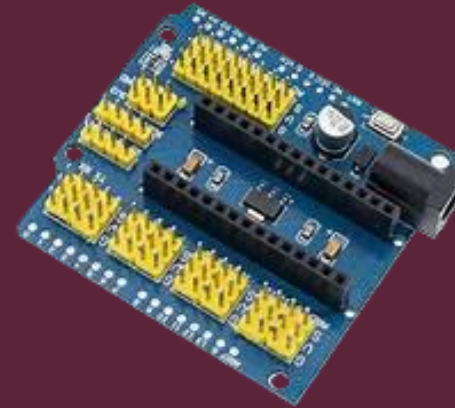


Core controller



Arduino Nano (CH340)

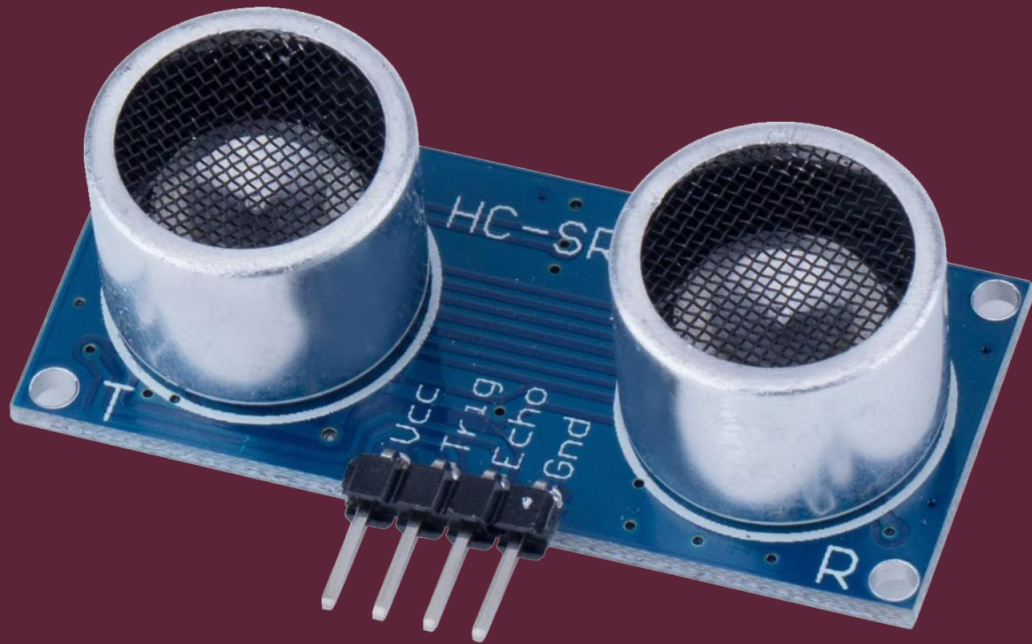
Program logic, PWM signals to servos.
Compact + breadboard-friendly.



Arduino Shield

Breakouts for servo headers, easier
wiring and power routing.

Sensors & Inputs



Ultrasonic Sensor (HC-SR04)

Distance sensing for obstacle detection and simple navigation.
Trig/echo to Nano.

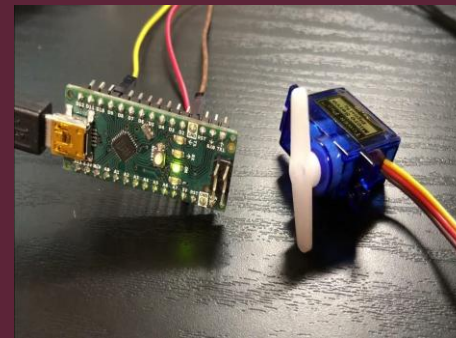
- ① Used for forward obstacle avoidance and basic environment awareness.

Actuation



8 × SG90 Servos

Leg joints (hip + knee). Lightweight and affordable. Controlled via PWM from Arduino.



Servo Testing

Individually tested angles to calibrate gait. Recorded neutral and travel limits for each leg.

Power System



+



USB charger

2pcs+1charger

Power Components

- 2 × 2500mA batteries — main energy source
- CC CV buck converter — steps voltage for servos/Arduino
- Capacitor 1000 µF — smooths servo current spikes
- On/off switch — safety and power cycling

□ Buck was added after initial failures — prevented early overheating of parts.

Why separate servo power from Arduino

Prevents voltage drops and resets. Isolates high current spikes from logic supply.

Cleaner logic

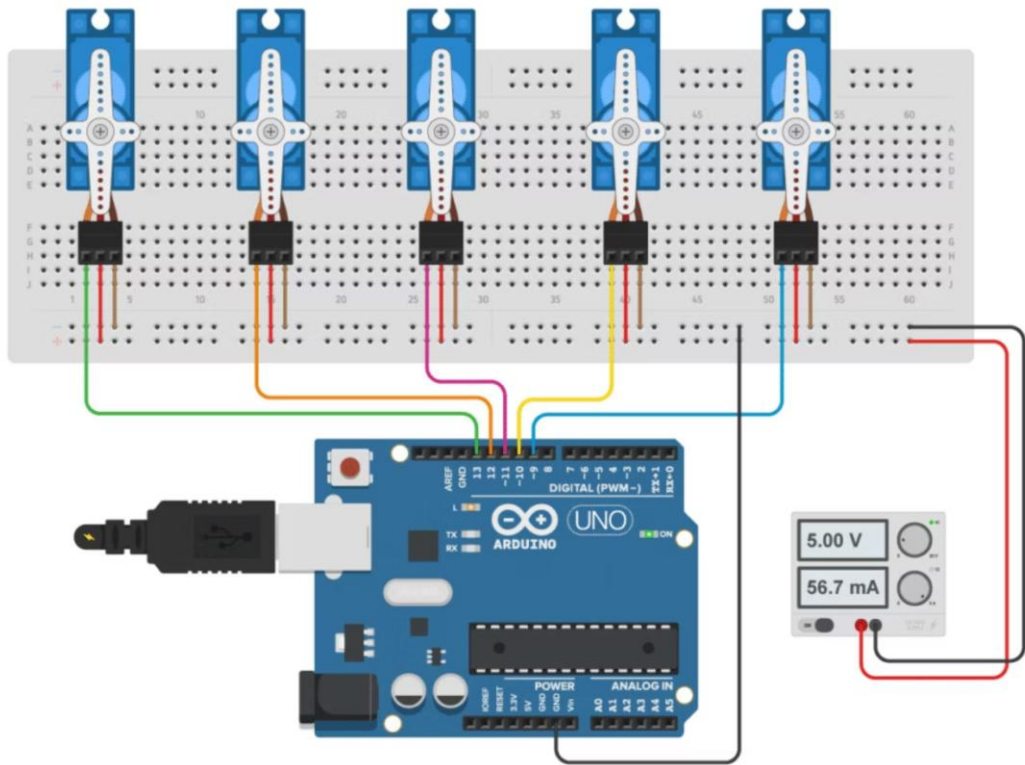
Arduino voltage stays stable → fewer brownouts.

Safer servos

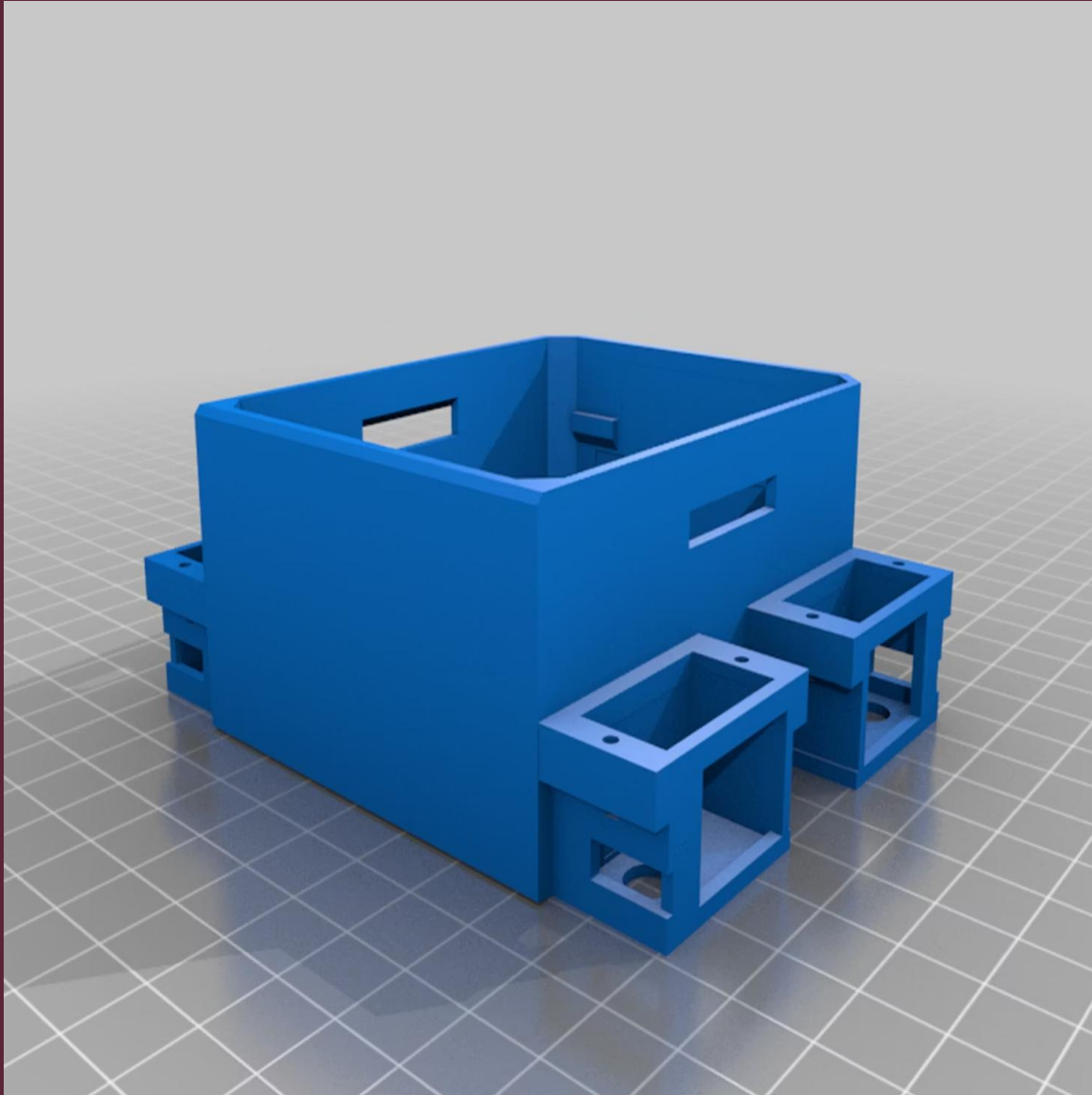
Servos draw high transient currents; dedicated supply + capacitor absorbs spikes.

Debugging

Easier to trace power faults when supplies are separated.

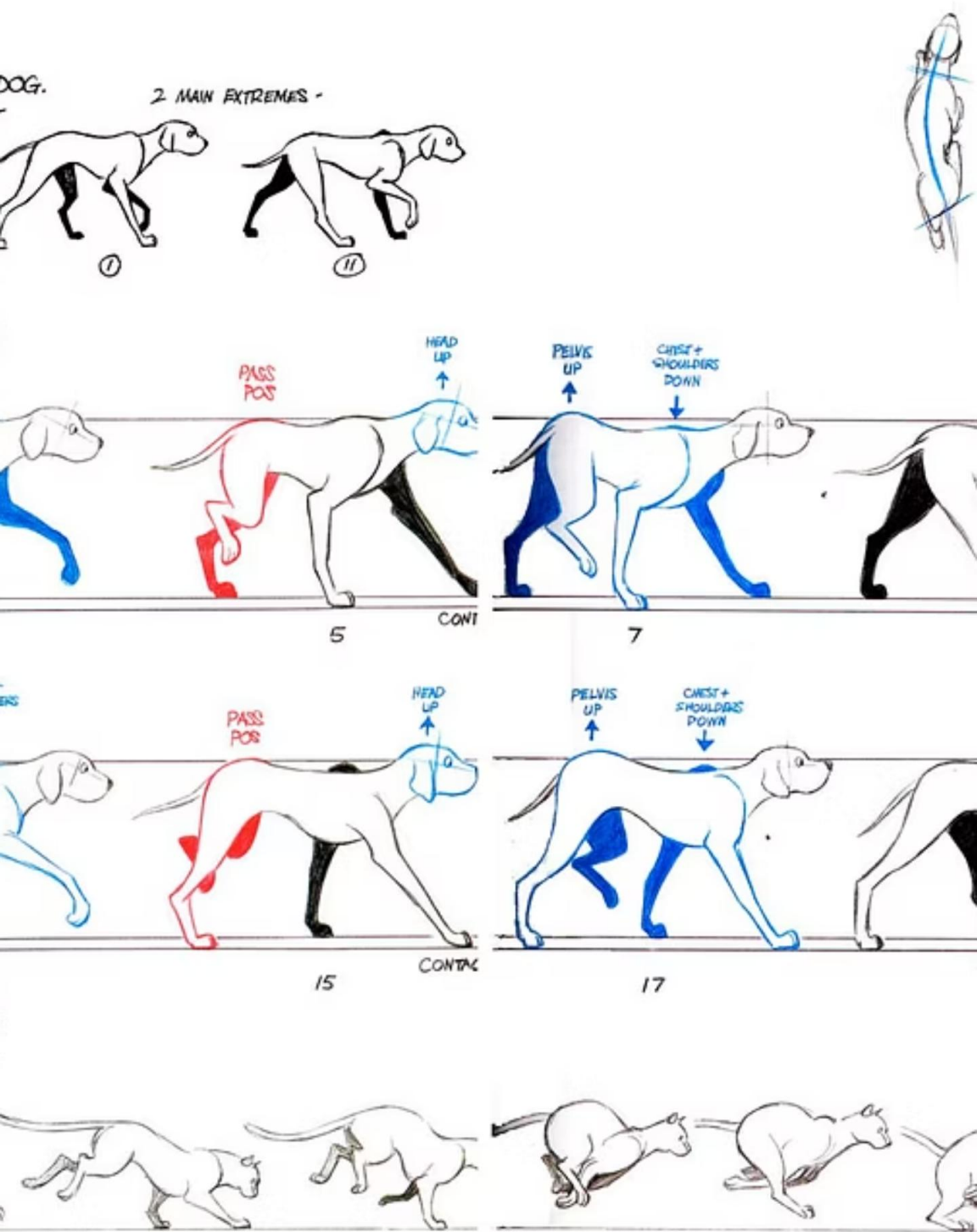


Manufacturing & Build Challenges

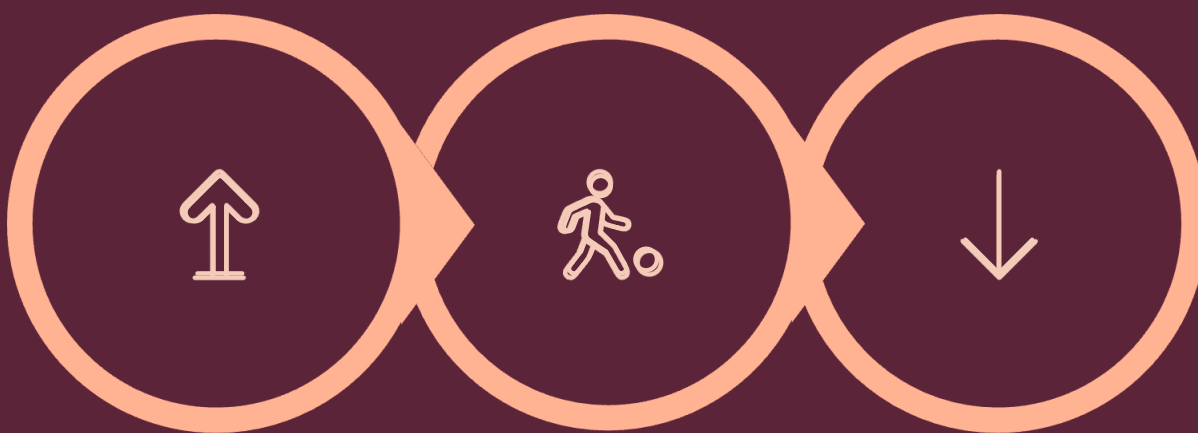


- 3D-print defects: warping, poor tolerances → misaligned joints
- Servo angle variation: required manual testing and trimming
- Early component failures: some parts burned before buck converter added

⚠ Quality of mounts directly affects gait and load distribution.



Motion & Gait



Lift

Swing

Place

Calibrated each servo angle and timing to create stable steps. Slow because motors under heavy load.

Improvements & Next Steps



Bluetooth Control

Remote commands and live tuning of gait parameters.



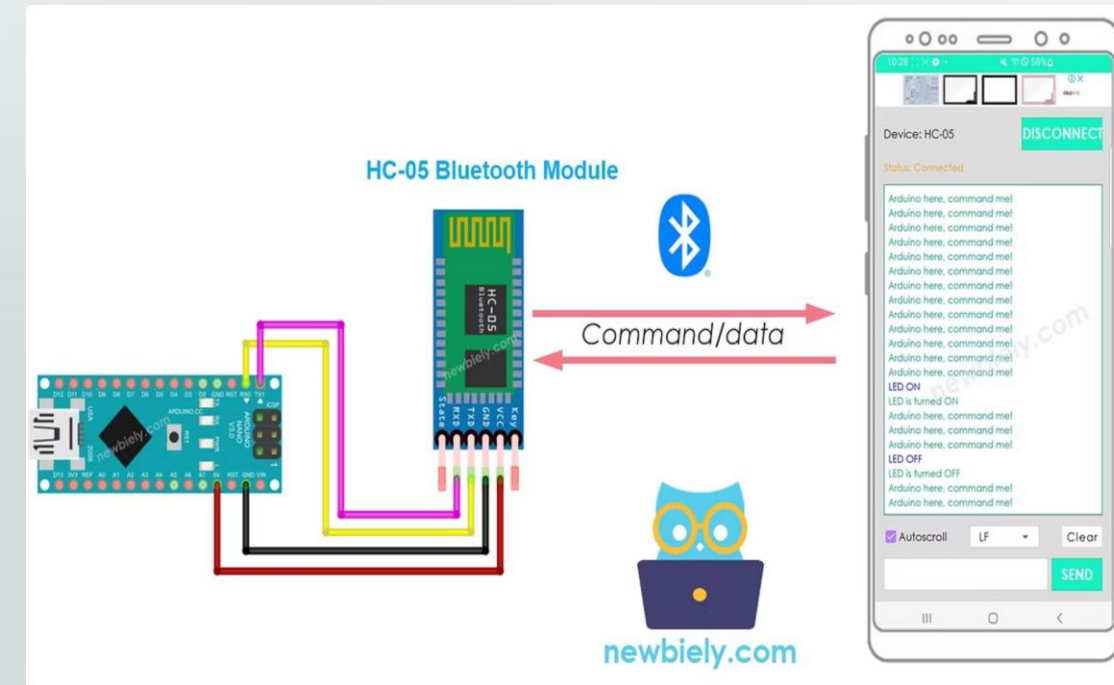
Weight reduction

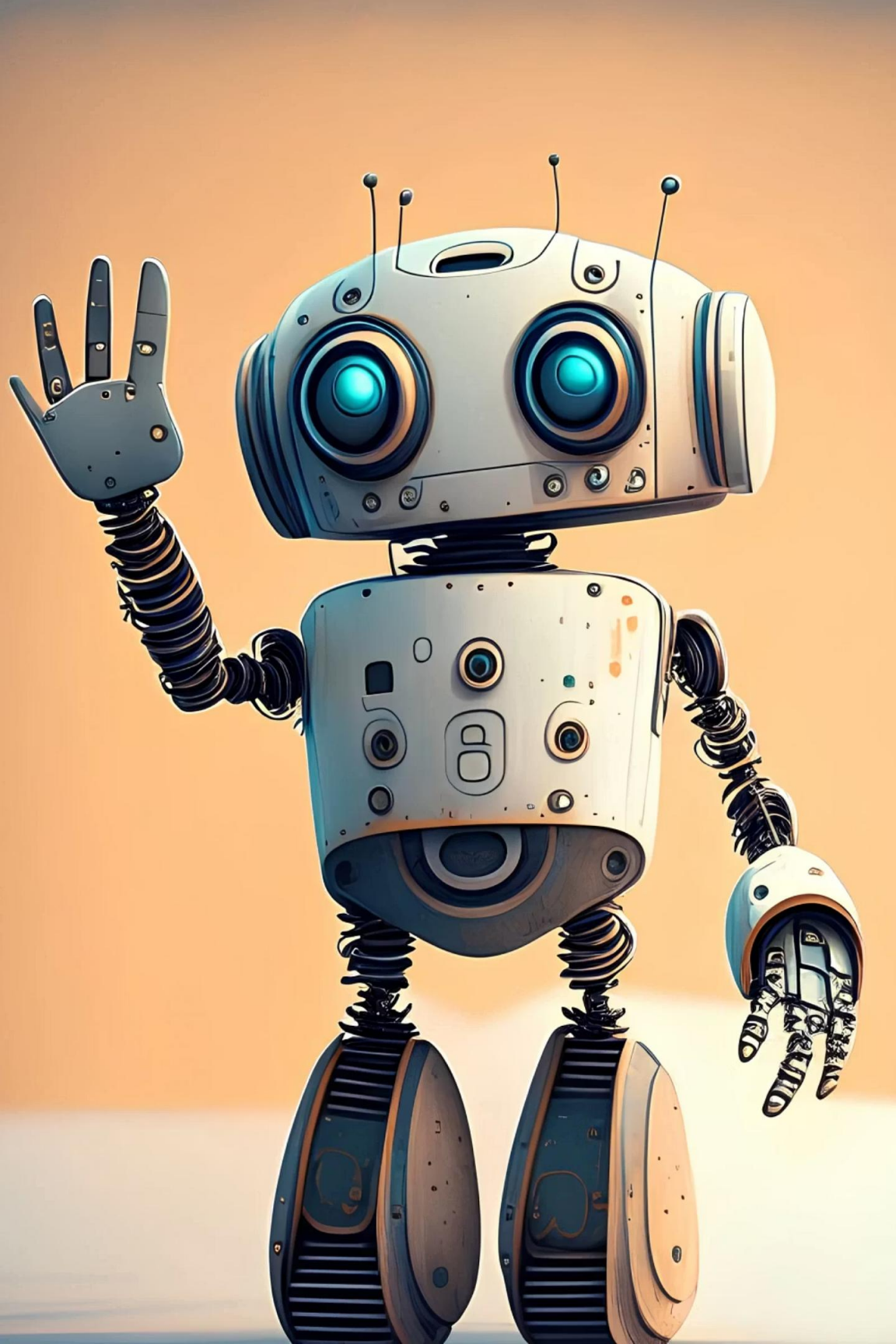
Lighter batteries or frame redesign to increase speed.



Mechanical improvements

Stronger 3D parts, better servo horns, optimized link lengths.





Key Takeaways

- Separate power: stable logic & reliable servos
- Test every servo angle — calibration is crucial
- Use a buck converter + capacitor to protect components
- Reduce weight and improve prints to raise speed
- Next: Bluetooth remote, refined gait, stronger parts

✓ Project is a solid learning platform — repeatable improvements will make Otto faster and more reliable.